MINISTRY OF EDUCATION, CULTURE AND RESEARCH
OF THE REPUBLIC OF MOLDOVA
Technical University of Moldova
Faculty of Computers, Informatics and Microelectronics
Department of Software Engineering and Automation

Bogdan Aramă FAF-242

Report

Laboratory work n.2

of Cryptography

Verified by:
Zaica Maia

Chisinau – 2026

1 Objective **3**
2 Short theory **3**
2.1 Monoalphabetic substitution 3
2.2 Frequency analysis 3
3 Implementation **4**
3.1 Letter Frequency Counting 4
3.2 Partial Substitution Key 5
3.3 Word and N-gram Patterns 5
3.4 Program Menu 6
4 Results and Verification **7**
4.1 The Intercepted Cryptogram 7
4.2 Letter Frequencies of the Cryptogram 8
4.3 Frequent Patterns 8
4.4 Naive Substitution by Rank 9
4.5 Breaking the Cipher Step by Step 10
4.6 Recovered Plaintext 12
4.7 Reconstructed Key 13
4.8 Manual Mode and Error Handling 13
5 Conclusions **14**
6 Source Code Repository **14**

Cryptanalysis of Monoalphabetic Ciphers

1 Objective

The objective is to break an intercepted cryptogram (Variant 1) that was produced by an unknown monoalphabetic substitution cipher, knowing only that the original message is written in English and that only the letters were encrypted, while digits, punctuation and spaces were left unchanged. The attack is based on frequency analysis: the letter frequencies of the cryptogram are compared with the known frequencies of the English language, and every guess is confirmed with frequent words, digraphs, trigraphs and the context of the message. A Java program was written to compute the statistics and to apply the substitutions step by step.

2 Short theory

2.1 Monoalphabetic substitution

A monoalphabetic cipher replaces every plaintext letter with one fixed ciphertext letter for the whole message. The key is a permutation of the alphabet, so for the 26-letter English alphabet the key space is:

|K| = 26! ≈ 4.03 × 10^26 possible keys

This makes an exhaustive key search impossible. Caesar, keyword and affine ciphers are all special cases of monoalphabetic substitution.

2.2 Frequency analysis

The weak point of the cipher is that the substitution does not hide the statistics of the language. If the plaintext letter E is encrypted as V, every E becomes V, so V will be about as frequent in the cryptogram as E is in English. For a long enough cryptogram, ordering its letters by frequency and comparing them with the English order gives the first candidate substitutions. Table 2.2 of the laboratory guide gives the English frequencies (%):

A	B	C	D	E	F	G	H	I	J	K	L	M
8.17	*1.49*	*2.78*	*4.25*	*12.70*	*2.23*	*2.01*	*6.09*	*6.97*	*0.15*	*0.77*	*4.03*	*2.41*
N	O	P	Q	R	S	T	U	V	W	X	Y	Z
6.75	*7.51*	*1.93*	*0.09*	*5.99*	*6.33*	*9.06*	*2.76*	*0.98*	*2.36*	*0.15*	*1.97*	*0.07*

Table 2.1. Letter frequencies of the English language

By frequency the English letters are ordered E T A O I N S H R D L C U M W F G Y P B V K J X Q Z. Frequencies alone are not enough, because a real text never matches the average exactly, so each guess is confirmed with other properties of the language:

- one-letter words: only „*a*” and „*I*”;
- most frequent words: *the, of, and, to, in, it, was, for*;
- most frequent digraphs: TH, HE, AN, IN, ER, ON, RE, ED, ND, HA, AT, EN;
- most frequent trigraphs: THE, AND, THA, ENT, ION, TIO, FOR, NDE, HAS, NCE, TIS, OFT, MEN;

- doubled letters: SS, EE, TT, OO, FF.

After a few substitutions, partially decrypted words appear (for example „*tQe*”) and are completed from context. This step needs human judgement, which is why the attack is carried out interactively.

3 Implementation

The program is written in Java (package **lab2**) and is structured into five classes:

Class	Responsibility
EnglishFrequency.java	English letter frequencies (table 2.1), digraphs, trigraphs, letter order by frequency
FrequencyAnalyzer.java	Counts the letters of the cryptogram, sorts them, finds frequent words, digraphs, trigraphs and doubled letters
SubstitutionKey.java	The partial key „cipher letter → plain letter”, the working view and the reconstructed alphabet
VariantOneAttack.java	The ten steps of the attack on Variant 1, each with its substitutions and reasoning
Main.java	Console menu, reading the ciphertext file, printing tables and results

3.1 Letter Frequency Counting

The analyzer counts only Latin letters, case-insensitively; digits, punctuation and spaces are skipped because they were not encrypted. The letters are then sorted by descending count, ties being broken alphabetically:

```
public FrequencyAnalyzer(String text) {
    this.text = text;
    for (char c : text.toCharArray()) {
        int i = EnglishFrequency.index(c);
        if (i >= 0) {
            counts[i]++;
            total++;
        }
    }
}
```

```
public String lettersByFrequency() {
    Character[] letters = new Character[EnglishFrequency.ALPHABET_SIZE];
    for (int i = 0; i < letters.length; i++) {
        letters[i] = EnglishFrequency.ALPHABET.charAt(i);
    }
    Arrays.sort(letters, (a, b) -> count(b) != count(a) ? count(b) - count(a) : a - b);

    StringBuilder sb = new StringBuilder();
    for (char c : letters) {
        sb.append(c);
    }
    return sb.toString();
}
```

3.2 Partial Substitution Key

The key stores, for each ciphertext letter, the guessed plaintext letter. Because a monoalphabetic key is a bijection, a guess that would give one plaintext letter two ciphertext images is rejected. The working view uses the notation of the laboratory guide: recovered letters in lowercase, still encrypted letters in UPPERCASE:

```
public void set(char cipher, char plain) {
    int ci = EnglishFrequency.index(cipher);
    int pi = EnglishFrequency.index(plain);
    if (ci < 0 || pi < 0) {
        throw new IllegalArgumentException("Both letters must be in A-Z.");
    }

    char p = Character.toLowerCase(plain);
    for (int i = 0; i < cipherToPlain.length; i++) {
        if (i != ci && cipherToPlain[i] == p) {
            throw new IllegalArgumentException("Plaintext '" + p + "' is already the image of ciphertext '"
                    + EnglishFrequency.ALPHABET.charAt(i) + "'. Remove that substitution first.");
        }
    }
    cipherToPlain[ci] = p;
}
```

```
public String apply(String cipherText) {
    StringBuilder sb = new StringBuilder();
    for (char c : cipherText.toUpperCase().toCharArray()) {
        int i = EnglishFrequency.index(c);
        if (i >= 0 && cipherToPlain[i] != UNKNOWN) {
            sb.append(cipherToPlain[i]);
        } else {
            sb.append(c);
        }
    }
    return sb.toString();
}
```

3.3 Word and N-gram Patterns

To confirm the guesses, the program finds the most frequent words of a given length and the most frequent n-grams. N-grams are counted inside words only, so that the spaces kept by the cipher are not crossed:

```
public List<Map.Entry<String, Integer>> topWords(int length, int limit) {
    Map<String, Integer> words = new HashMap<>();
    for (String w : text.toUpperCase().split("[^A-Z]+")) {
        if (w.length() == length) {
            words.merge(w, 1, Integer::sum);
        }
    }
    return top(words, limit);
}
```

```
public List<Map.Entry<String, Integer>> topNgrams(int n, int limit) {
    Map<String, Integer> grams = new HashMap<>();
    for (String w : text.toUpperCase().split("[^A-Z]+")) {
        for (int i = 0; i + n <= w.length(); i++) {
            grams.merge(w.substring(i, i + n), 1, Integer::sum);
        }
    }
    return top(grams, limit);
}
```

3.4 Program Menu

Options 5–7 let the analyst carry out the attack by hand, in the same way as the online service crypto.interactive-maths.com. Option 8 replays the attack described in section 4.5, one step per Enter key. Option 9 loads any other ciphertext file, so the tool is not tied to Variant 1. Each step of the replay is stored as a record with its substitutions and reasoning:

```
Select Action:
1. Show ciphertext
2. Letter frequency analysis (vs. English, table 2.2)
3. Frequent words, digraphs, trigraphs and doubles
4. Naive substitution by frequency rank
5. Add substitution (e.g. W=t)
6. Remove substitution
7. Show current decryption and key
8. Replay the Variant 1 attack step by step
9. Load ciphertext from file
10. Reset key
0. Exit
Choice: 0
```

```
public record Step(String substitutions, String reasoning) {}

public static final List<Step> STEPS = List.of(
    new Step("V=e W=t",
        "The most frequent ciphertext letters are V (12.74%) and W (10.75%). In English the most frequent "
        + "letters are E (12.70%) and T (9.06%), so we assume V -> e and W -> t."),
    new Step("Q=h",
        "The 3-letter word 'tQe' appears 55 times. The most common English word is 'the', so Q -> h. "
        + "'th' and 'he' are also the two most frequent English digraphs."),
    ...
```

4 Results and Verification

4.1 The Intercepted Cryptogram

The cryptogram of Variant 1 (stored in **lab2/ciphertext_v1.txt**) is:

> *NG T OTF gvtisf 4,000 fvtip tjn, xg t wnrg htssvo Zvgvw Lqdcdaniovixgj wqv wqxg ixaang nc wqv Gxsv, t ztpwvi phixav plvwhqvo ndw wqvqxvinjsfuqp wqtw wnso wqv pwnif nc qxp snio'p sxcv— tgo xg pn onxgj qvnuvgvo wqv ivhniovo qxpwnif nc hifuwnsnjf. Qxp rtp gnw t pfpwvz nc pvhivwrixwxgj tp wqv znovig rniso lgnrp xw; qv dpvo gn cdssf ovkvsnuvo hnov nc qxvinjsfuqxhpfzans pdapwxwdwxngp. Qxp xgphixuwxng, htikvo tandw 1900 A.H. xgwn wqvsxkxgj inhl xg wqv ztxg hqtzavi nc wqv wnza nc wqv gnasvztgLqgdzqnwvu XX, zvivsf dpvp pnzv dgdpdts qxvinjsfuqxh pfzansp qvivtgo wqviv xg usthv nc wqv zniv nioxgtif ngvp. Znpw nhhdi xg wqv stpw 20hnsdzgp nc wqv xgphixuwxng'p 222, xg t pvhwxng ivhnioxgj wqv zngdzvgwpwqtw Lqgdzqnwvu qto vivhwvo xg wqv pvikxhv nc wqv uqtitnq TzvgvzqvwXX. Wqv xgwvgwxng rtp gnw wn ztlv xw qtio wn ivto wqv wvyw. Xw rtp wnxzutiw t oxjgxwf tgo tdwqnixwf wn xw, uviqtup xg wqv ptzv rtf wqtw tjnkvigzvgw uinhstztwxng rxss puvss ndw "Xg wqv fvti nc Ndi Snio Ngvwqndptgo vxjqw qdgoivo tgo pxywf wqivv" xgpwvto nc edpw rixwxgj "1863."Wqv tgngfzndp phixav ztf tspn qtkv avvg ovzngpwitwxgj qxp lgnrsvojvcni unpwvixwf. Wqdp wqv xgphixuwxng rtp gnw pvhivw rixwxgj, adw xwxghniunitwvo ngv nc wqv vppvgwxts vsvzvgwp nc hifuwnjituqf: t ovsxavitwvwitgpcniztwxng nc wqv rixwxgj. Xw xp wqv nsovpw wvyw lgnrg wn on pn.Wqv witgpcniztwxngp nhhdi xg cdgvitif cnizdstp, xg t qfzg wn Wqnwq,xg t hqtuwvi nc wqv Annl nc wqv Ovto, ng wqv ptihnuqtjdp nc wqv uqtitnqPvwx X, xg infts wxwsvp oxpustfvo xg Sdyni, ng wqv tihqxwitkv nc wqv Wvzusv nc Sdyni, ng pwvsv, xgstdotwnif axnjituqxh xgphixuwxngp. Wqviv xp gnwqxgj zvtgw wn avhnghvtsvo xg tss wqxp; xgovvo, ztgf nc wqv pwtwvzvgwp tiv ivuvtwvo xgnioxgtif cniz ixjqw gvyw wn wqv tswvivo ngvp. Rqf, wqvg, wqvwitgpcniztwxngp? Pnzvwxzvp cni vppvgwxtssf wqv ptzv ivtpng tp xgLqgdzqnwvu'p wnza: wn xzuivpp wqv ivtovi. Nhhtpxngtssf cni thtssxjituqxh ni ovhnitwxkv vccvhw; itivsf, wn xgoxhtwv t hngwvzunitifuingdghxtwxng; uviqtup vkvg cni t ovsxavitwv tihqtxpz tp t ivthwxngtjtxgpw cnivxjg xgcsdvghv.Adw ztgf xgphixuwxngp tiv wxghwdivo, cni wqv cxipw wxzv, rxwq wqvpvhngo vppvgwxts cni hifuwnsnjf—pvhivhf. Xg t cvr htpvp, wqv pvhivhf rtpxgwvgovo wn xghivtpv wqv zfpwvif tgo qvghv wqv tihtgv ztjxhts unrvip nchviwtxg ivsxjxndp wvywp. Adw wqv pvhivhf xg ztgf zniv htpvp ivpdswvo cinzwqv dgovipwtgotasv ovpxiv nc wqv Vjfuwxtgp wn qtkv utppvipaf ivto wqvxivuxwtuqp tgo pn hngcvi dung wqv ovutiwvo wqv asvppxgjp rixwwvg wqvivxg.Xg Vjfuw, rxwq xwp hnghvgwitwxng dung wqv tcwvisxcv, wqv gdzavi nc wqvpvxgphixuwxngp pnng • uinsxcvitwvo wn pdhq tg vywvgw wqtw wqv twwvgwxng tgowqv jnnorxss nc kxpxwnip cstjjvo. Wn ivkxkv wqvxi xgwvivpw, wqv phixavpovsxavitwvsf ztov wqv xgphixuwxngp t axw naphdiv. Wqvf xgwinodhvo wqvhifuwnjituqxh pxjgp wn htwhq wqv ivtovi'p vfv, ztlv qxz rngovi, tgowvzuw qxz xgwn dgixoosxgj wqvz — tgo pn xgwn ivtoxgj wqv asvppxgjp. Xwrtp t pniw nc Ztoxpng Tkvgdv wvhqgxbdv xg wqv Ktssvf nc wqv Lxgjp. Adwwqv wvhqgxbdv ctxsvo dwwvisf. Xgpwvto nc xgwvivpwxgj wqv ivtovip, xwvkxovgwsf ovpwinfvo vkvg wqv psxjqwvpw ovpxiv wn ivto wqv vuxwtuqp, cnipnng tcwvi wqv cdgvitif hifuwnjituqf rtp avjdg, xw rtp tatgongvo.*

4.2 Letter Frequencies of the Cryptogram

The first step is to count all letters of the cryptogram (option 2). Both columns are sorted by frequency, so every row pairs a ciphertext letter with the English letter of the same rank:

```
Choice: 2

--- Letter frequencies (2567 letters) ---
Rank  Cipher  Count     %   Histogram                 | English     %   Histogram
  1     V     327   12.74   ########################  |    E    12.70   ########################
  2     W     276   10.75   ####################      |    T     9.06   ################
  3     N     205    7.99   ###############           |    A     8.17   ###############
  4     X     198    7.71   ###############           |    O     7.51   ##############
  5     G     196    7.64   ###############           |    I     6.97   #############
  6     T     178    6.93   #############             |    N     6.75   #############
  7     I     176    6.86   #############             |    S     6.33   ############
  8     P     165    6.43   ############              |    H     6.09   ############
  9     Q     145    5.65   ##########                |    R     5.99   ############
 10     O      93    3.62   #######                   |    D     4.25   ########
 11     H      83    3.23   ######                    |    L     4.03   ########
 12     S      80    3.12   ######                    |    C     2.78   ######
 13     Z      63    2.45   #####                     |    U     2.76   ######
 14     F      61    2.38   #####                     |    M     2.41   #####
 15     C      59    2.30   #####                     |    W     2.36   #####
 16     U      58    2.26   #####                     |    F     2.23   ####
 17     D      56    2.18   ####                      |    G     2.01   ####
 18     J      45    1.75   ####                      |    Y     1.97   ####
 19     A      35    1.36   ###                       |    P     1.93   ####
 20     R      27    1.05   ##                        |    B     1.49   ###
 21     K      17    0.66   #                         |    V     0.98   ##
 22     L      13    0.51   #                         |    K     0.77   ##
 23     Y       8    0.31   #                         |    J     0.15
 24     B       2    0.08                             |    X     0.15
 25     E       1    0.04                             |    Q     0.09
 26     M       0    0.00                             |    Z     0.07

Ciphertext letters by frequency : VWNXGTIPQOHSZFCUDJARKLYBEM
English letters by frequency    : ETAOINSHRDLCUMWFGYPBVKJXQZ
```

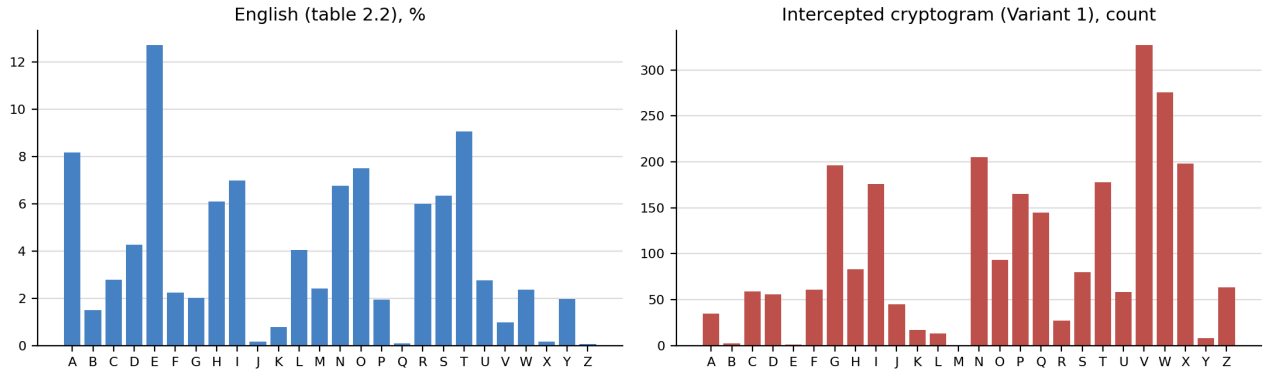


Figure 4.1. Letter frequencies of English (left) and of the intercepted cryptogram (right)

4.3 Frequent Patterns

Option 3 lists the most frequent short words, digraphs, trigraphs and doubled letters of the cryptogram:

```
Choice: 3

--- Frequent patterns in the ciphertext ---
1-letter words : T(15), P(4), A(1), H(1), X(1)
2-letter words : NC(27), XG(19), WN(15), XW(6), NG(4), PN(4), TP(3), XP(2)
3-letter words : WQV(55), RTP(6), TGO(6), CNI(5), QXP(4), ADW(3), GNW(3), NDW(2)
4-letter words : IVTO(3), WQTW(3), XGWN(3), ZTGF(3), DUNG(2), NGVP(2), PTZV(2), QTKV(2)
Digraphs       : WQ(87), QV(78), XG(68), NG(45), IV(44), WV(43), VI(42), WX(37), NI(36), WN(32),
NC(28), XW(27)
Trigraphs      : WQV(72), XNG(19), WXN(18), XGJ(16), CNI(13), TGO(12), VGW(12), IVT(11), HIX(10),
PHI(10), VTO(10), WVI(10)
Double letters : SS(10), PP(7), NN(4), WW(4), HH(3), VV(3)

English digraphs  : TH HE AN IN ER ON RE ED ND HA AT EN
English trigraphs : THE AND THA ENT ION TIO FOR NDE HAS NCE TIS OFT MEN
English doubles   : SS EE TT OO FF
```

The 3-letter word WQV (55 times) and the trigraph WQV (72 times) stand out exactly like „*the*" in English.

4.4 Naive Substitution by Rank

If every ciphertext letter is replaced by the English letter with the same rank (option 4), the result is unreadable, because N, X, G, T and I all have frequencies between 6.9% and 8.0% and their order differs from English. Only V → e and W → t are correct:

```
Choice: 4

--- Naive substitution by frequency rank ---
Cipher  : VWNXGTIPQOHSZFCUDJARKLYBEM
English : etaoinshrdlcumwfgypbvkjxqz

ai n dnm ienscm 4,000 mensh nya, oi n tabi lncced ueiet krgwgpasdesoiy tre troi soppai aw tre ioce,
n unhtes hlsope hketlred agt treroesaycmfrh trnt tacd tre htasm aw roh casd'h cowe— nid oi ha daoiy
reafeied tre selasded rohtasm aw lsmftacaym. roh bnh iat n hmhteu aw helsetbsotoiy nh tre uadesi
bascd kiabh ot; re ghed ia wgccm devecafed lade aw roesaycmfrolhmupac hgphtotgtoaih. roh
oihlsoftoai, l ...

[Info] The result is unreadable: real texts deviate from the average frequencies,
       so each guess has to be confirmed with words, digraphs and context.
```

4.5 Breaking the Cipher Step by Step

The attack is replayed with option 8. Each step below shows the reasoning and the beginning of the partially decrypted text (lowercase = recovered, UPPERCASE = still encrypted). The first step as printed by the program:

```
Press Enter for step 1/10 (or q to stop)...

=== Step 1: V -> e W -> t ===
The most frequent ciphertext letters are V (12.74%) and W (10.75%). In English the most frequent
letters are E (12.70%) and T (9.06%), so we assume V -> e and W -> t.

NG T OTF GeTISF 4,000 FeTIP TJN, XG T tNRG HTSSeO ZeGet LQDCDANIOeIXGJ tQe tQXG IXAANG NC tQe GXSe,
T ZTPteI PHIXAe PLetHQeO NDt tQeQXeINJSFUQP tQTt tNSO tQe PtNIF NC QXP SNIO'P SXCe— TGO XG PN ONXGJ
QeNUeGeO tQe IeHNIOeO QXPtNIF NC HIFUtNSNJF. QXP RTP GNt T PFPteZ NC PeHIetRIXtXGJ TP tQe ZNOeIG
RNISO LGNRP Xt; Qe DPeO GN CDSSF OeKeSNUeO HNOe NC QXeINJSFUQXHPFZANS PDAPtXtDtXNGP. QXP
XGPHIXUtXNG, HTIKeO TANDt 1900 A.H. XGtN tQeSXKXGJ INHL XG tQe ZTXG HQTZAeI NC tQe tNZA NC tQe
GNASeZTGLQGDZQNteU XX, ZeIeSF DPeP PNZe ...
```

Step 1: V → e, W → t. The two most frequent ciphertext letters, **V** (12.74%) and **W** (10.75%), stand out from the rest exactly like **E** (12.70%) and **T** (9.06%) in English, so we assume V → e and W → t:

> *NG T OTF GeTISF 4,000 FeTIP TJN, XG T tNRG HTSSeO ZeGet LQDCDANIOeIXGJ tQe tQXG IXAANG NC tQe GXSe, T ZTPteI PHIXAe PLetHQeO NDt tQeQXeINJSFUQP tQTt tNSO tQe PtNIF NC QXP SNIO'P SXCe— TGO XG PN ONXGJ QeNUeGeO tQe IeHNIOeO QXPtNIF NC HIFUtNSNJF. QXP RTP GNt T PFPteZ NC PeHIetRIXtXGJ TP tQe ZNOeIG RNISO LGNRP Xt; Qe DPeO GN CDSSF OeKeSNUeO HNOe NC QXeINJSFUQXHPFZANS PDAPtXtDtXNGP. QXP XGPHIXUtXNG, HTIKeO TANDt 1900 A.H. XGtN ...*

Step 2: Q → h. The word „*tQe*" appears 55 times. The most frequent English word is „*the*", so Q → h. This agrees with the digraphs: WQ (87) and QV (78) are the most frequent ciphertext digraphs, while TH and HE are the most frequent English ones:

> *NG T OTF GeTISF 4,000 FeTIP TJN, XG T tNRG HTSSeO ZeGet LhDCDANIOeIXGJ the thXG IXAANG NC the GXSe, T ZTPteI PHIXAe PLetHheO NDt thehXeINJSFUhP thTt tNSO the PtNIF NC hXP SNIO'P SXCe— TGO XG PN ONXGJ heNUeGeO the IeHNIOeO hXPtNIF NC HIFUtNSNJF. hXP RTP GNt T PFPteZ NC PeHIetRIXtXGJ TP the ZNOeIG RNISO LGNRP Xt; he DPeO GN CDSSF OeKeSNUeO HNOe NC hXeINJSFUhXHPFZANS PDAPtXtDtXNGP. hXP XGPHIXUtXNG, HTIKeO TANDt 1900 A.H. XGtN ...*

Step 3: T → a. „*T*" appears 15 times as a word by itself. The only one-letter English words are „*a*" and „*I*", and T has 6.93% (A has 8.17%), so T → a. The word „*thTt*" now reads „*that*":

> *NG a OaF GeaISF 4,000 FeaIP aJN, XG a tNRG HaSSeO ZeGet LhDCDANIOeIXGJ the thXG IXAANG NC the GXSe, a ZaPteI PHIXAe PLetHheO NDt thehXeINJSFUhP that tNSO the PtNIF NC hXP SNIO'P SXCe— aGO XG PN ONXGJ heNUeGeO the IeHNIOeO hXPtNIF NC HIFUtNSNJF. hXP RaP GNt a PFPteZ NC PeHIetRIXtXGJ aP the ZNOeIG RNISO LGNRP Xt; he DPeO GN CDSSF OeKeSNUeO HNOe NC hXeINJSFUhXHPFZANS PDAPtXtDtXNGP. hXP XGPHIXUtXNG, HaIKeO aANDt 1900 A.H. XGtN ...*

Step 4: N → o, C → f. The word „*tN*" (15 times) can only be „*to*", so N → o (7.99% against 7.51% for O). The most frequent two-letter word „*oC*" (27 times) is then „*of*", so C → f:

> *oG a OaF GeaISF 4,000 FeaIP aJo, XG a toRG HaSSeO ZeGet LhDfDAoIOeIXGJ the thXG IXAAoG of the GXSe, a ZaPteI PHIXAe PLetHheO oDt thehXeIoJSFUhP that toSO the PtoIF of hXP SoIO'P SXfe— aGO XG Po OoXGJ heoUeGeO the IeHoIOeO hXPtoIF of HIFUtoSoJF. hXP RaP Got a PFPteZ of PeHIetRIXtXGJ aP the ZoOeIG RoISO LGoRP Xt; he DPeO Go fDSSF OeKeSoUeO HoOe of hXeIoJSFUhXHPFZAoS PDAPtXtDtXoGP. hXP XGPHIXUtXoG, HaIKeO aAoDt 1900 A.H. XGto ...*

Step 5: G → n, O → d. „*aGO*" appears 6 times. After „*the*", the most frequent three-letter English word is „*and*", so G → n and O → d:

> *on a daF neaISF 4,000 FeaIP aJo, Xn a toRn HaSSed Zenet LhDfDAoIdeIXnJ the thXn IXAAon of the nXSe, a ZaPteI PHIXAe PLetHhed oDt thehXeIoJSFUhP that toSd the PtoIF of hXP SoId'P SXfe— and Xn Po doXnJ heoUened the IeHoIded hXPtoIF of HIFUtoSoJF. hXP RaP not a PFPteZ of PeHIetRIXtXnJ aP the ZodeIn RoISd LnoRP Xt; he DPed no fDSSF deKeSoUed Hode of hXeIoJSFUhXHPFZAoS PDAPtXtDtXonP. hXP XnPHIXUtXon, HaIKed aAoDt 1900 A.H. Xnto ...*

Step 6: X → i. „*Xn*" (19 times) and „*Xt*" (6 times) are „*in*" and „*it*". X has 7.71%, close to I (6.97%):

> *on a daF neaISF 4,000 FeaIP aJo, in a toRn HaSSed Zenet LhDfDAoIdeIinJ the thin IiAAon of the niSe, a ZaPteI PHIiAe PLetHhed oDt thehieIoJSFUhP that toSd the PtoIF of hiP SoId'P Sife— and in Po doinJ heoUened the IeHoIded hiPtoIF of HIFUtoSoJF. hiP RaP not a PFPteZ of PeHIetRIitinJ aP the ZodeIn RoISd LnoRP it; he DPed no fDSSF deKeSoUed Hode of hieIoJSFUhiHPFZAoS PDAPtitDtionP. hiP inPHIiUtion, HaIKed aAoDt 1900 A.H. into ...*

Step 7: I → r, P → s, R → w. „*foI*" is „*for*", so I → r. „*hiP*" is „*his*", so P → s. Then the frequent word „*Ras*" (6 times) can only be „*was*", so R → w:

> *on a daF nearSF 4,000 Fears aJo, in a town HaSSed Zenet LhDfDAorderinJ the thin riAAon of the niSe, a Zaster sHriAe sLetHhed oDt thehieroJSFUhs that toSd the storF of his Sord's Sife— and in so doinJ heoUened the reHorded historF of HrFUtoSoJF. his was not a sFsteZ of seHretwritinJ as the Zodern worSd Lnows it; he Dsed no fDSSF deKeSoUed Hode of hieroJSFUhiHsFZAoS sDAstitDtions. his insHriUtion, HarKed aAoDt 1900 A.H. into ...*

Step 8: H → c, F → y, U → p, S → l, J → g. In the phrase „*the reHorded historF of HrFUtoSoJF*" the 10-letter word „*HrFUtoSoJF*" can only be „*cryptology*" (the text is about its history). This one word gives five letters and also completes „*nearly*", „*years*", „*ago*", „*called*", „*told*" and „*story*":

> *on a day nearly 4,000 years ago, in a town called Zenet LhDfDAordering the thin riAAon of the nile, a Zaster scriAe sLetched oDt thehieroglyphs that told the story of his lord's life— and in so doing heopened the recorded history of cryptology. his was not a systeZ of secretwriting as the Zodern world Lnows it; he Dsed no fDlly deKeloped code of hieroglyphicsyZAol sDAstitDtions. his inscription, carKed aAoDt 1900 A.c. into ...*

Step 9: A → b, D → u, Z → m. „*ADt*” (3 times) is „*but*”, so A → b and D → u. „*Zany*” (3 times) is „*many*”, so Z → m, which also confirms „*ribbon*”, „*scribe*”, „*master*” and „*modern*”:

> *on a day nearly 4,000 years ago, in a town called menet Lhufubordering the thin ribbon of the nile, a master scribe sLetched out thehieroglyphs that told the story of his lord's life— and in so doing heopened the recorded history of cryptology. his was not a system of secretwriting as the modern world Lnows it; he used no fully deKeloped code of hieroglyphicsymbol substitutions. his inscription, carKed about 1900 b.c. into ...*

Step 10: K → v, L → k, Y → x, B → q, E → j. The remaining rare letters come from context: „*deKeloped*” → „*developed*”, „*Lnows*” → „*knows*”, „*teYt*” → „*text*”, „*techniBue*” → „*technique*”, „*Eust*” → „*just*”:

> *on a day nearly 4,000 years ago, in a town called menet khufubordering the thin ribbon of the nile, a master scribe sketched out thehieroglyphs that told the story of his lord's life— and in so doing heopened the recorded history of cryptology. his was not a system of secretwriting as the modern world knows it; he used no fully developed code of hieroglyphicsymbol substitutions. his inscription, carved about 1900 b.c. into ...*

Ciphertext letter **M** never appears in the cryptogram: it is the only unused ciphertext letter and „*z*” is the only plaintext letter left, so by elimination z → M.

4.6 Recovered Plaintext

> *ON A DAY nearly 4,000 years ago, in a town called Menet Khufubordering the thin ribbon of the Nile, a master scribe sketched out thehieroglyphs that told the story of his lord's life— and in so doing heopened the recorded history of cryptology. His was not a system of secretwriting as the modern world knows it; he used no fully developed code of hieroglyphicsymbol substitutions. His inscription, carved about 1900 B.C. into theliving rock in the main chamber of the tomb of the noblemanKhnumhotep II, merely uses some unusual hieroglyphic symbols hereand there in place of the more ordinary ones. Most occur in the last 20columns of the inscription's 222, in a section recording the monumentsthat Khnumhotep had erected in the service of the pharaoh AmenemhetII. The intention was not to make it hard to read the text. It was toimpart a dignity and authority to it, perhaps in the same way that agovernment proclamation will spell out "In the year of Our Lord Onethousand eight hundred and sixty three" instead of just writing "1863."The anonymous scribe may also have been demonstrating his knowledgefor posterity. Thus the inscription was not secret writing, but itincorporated one of the essential elements of cryptography: a deliberatetransformation of the writing. It is the oldest text known to do so.The transformations occur in funerary formulas, in a hymn to Thoth,in a chapter of the Book of the Dead, on the sarcophagus of the pharaohSeti I, in royal titles displayed in Luxor, on the architrave of the Temple of Luxor, on stele, inlaudatory biographic inscriptions. There is nothing meant to beconcealed in all this; indeed, many of the statements are repeated inordinary form right next to the altered ones. Why, then, thetransformations? Sometimes for essentially the same reason as inKhnumhotep's tomb: to impress the reader. Occasionally for acalligraphic or decorative effect; rarely, to indicate a contemporarypronunciation; perhaps even for a deliberate archaism as a reactionagainst foreign influence.But many inscriptions are tinctured, for the first time, with thesecond essential for cryptology—secrecy. In a few cases, the secrecy wasintended to increase the mystery and hence the arcane magical powers ofcertain religious texts. But the secrecy in many more cases resulted fromthe understandable desire of the Egyptians to have passersby read theirepitaphs and so confer upon the departed the blessings written therein.In Egypt, with its concentration upon the afterlife, the number of theseinscriptions soon • proliferated to such an*

extent that the attention andthe goodwill of visitors flagged. To revive their interest, the scribesdeliberately made the inscriptions a bit obscure. They introduced thecryptographic signs to catch the reader's eye, make him wonder, andtempt him into unriddling them — and so into reading the blessings. Itwas a sort of Madison Avenue technique in the Valley of the Kings. Butthe technique failed utterly. Instead of interesting the readers, itevidently destroyed even the slightest desire to read the epitaphs, forsoon after the funerary cryptography was begun, it was abandoned.

Some words are joined (e.g. „*thehieroglyphs*") because the spaces were already missing at the line breaks of the ciphertext given in the assignment; the cipher does not change spaces. The text is an excerpt from David Kahn's *The Codebreakers* about the inscription in the tomb of Khnumhotep II (about 1900 B.C.), the oldest known deliberate transformation of writing.

4.7 Reconstructed Key

Placing the plaintext alphabet above the ciphertext letters gives the key of the cryptogram (option 7):

Plaintext	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
Ciphertext	T	A	H	O	V	C	J	Q	X	E	L	S	Z	G	N	U	B	I	P	W	D	K	R	Y	F	M

Table 4.1. Reconstructed alphabet of the intercepted message

```
Reconstructed alphabet (table 2.5):
Plaintext  : a b c d e f g h i j k l m n o p q r s t u v w x y z
Ciphertext : T A H O V C J Q X E L S Z G N U B I P W D K R Y F -
```

The key alphabet is regular: each next ciphertext letter is 7 positions after the previous one (T, A, H, O, V, C, J, ...). With a = 0, ..., z = 25 the key is the affine cipher

- **Encryption: c = (7 · p + 19) mod 26**
- **Decryption: p = 15 · (c − 19) mod 26 (15 = 7^-1 mod 26)**

An affine cipher is a special case of monoalphabetic substitution, so frequency analysis breaks it in the same way. Other messages from the same sender could now be decrypted directly with this formula.

4.8 Manual Mode and Error Handling

In manual mode the analyst enters guesses such as **V=e**. A guess that would map two ciphertext letters to the same plaintext letter, or a malformed pair, is rejected, and substitutions can be removed again:

```
Choice: 5
Enter substitutions CIPHER=plain separated by spaces (e.g. V=e W=t): V=e W=e q=1 X=i
[OK] V -> e
[Error] Plaintext 'e' is already the image of ciphertext 'V'. Remove that substitution first.
[Error] 'q=1' is not in the form X=y.
[OK] X -> i

NG T OTF GeTISF 4,000 FeTIP TJN, iG T WNRG HTSSeO ZeGeW LQDCDANIOeIiGJ WQe WQiG IiAANG NC WQe GiSe,
T ZTPWeI PHIiAe PLeWHQeO NDW WQeQieINJSFUQP WQTW WNSO WQe PWNIF NC QiP SNIO'P SiCe— TGO iG PN ONiGJ
QeNUeGeO WQe IeHNIOeO QiPWNIF NC HIFUWNSNJF. QiP RTP GNW T PFPWeZ NC PeHIeWRIiWiGJ TP WQe ZNOeIG
RNISO LGNRP iW; Qe DPeO GN CDSSF OeKeSNUeO HNOe NC QieINJSFUQiHPFZANS PDAPWiWDWiNGP. QiP
iGPHIiUWiNG, H ...

Choice: 6
Enter ciphertext letter to remove: Q
[Error] Letter Q has no substitution.

Choice: 6
Enter ciphertext letter to remove: X
[OK] Removed substitution for X.
```

5 Conclusions

In this laboratory work a cryptogram encrypted with an unknown monoalphabetic cipher was broken using frequency analysis, with the help of a Java tool for statistics and step-by-step substitution. The key findings are:

1. **Large key space is not security:** a monoalphabetic substitution has 26! ≈ 4 × 10^26 keys, yet the 2567-letter cryptogram was broken in ten steps without trying any key, because the cipher preserves the letter statistics of the plaintext.
2. **Frequencies give a starting point, not the answer:** the two most frequent letters matched E and T exactly, but substitution by rank alone gave an unreadable text, since letters with close frequencies change places.
3. **Words, digraphs and context confirm the guesses:** „*the*” (55 times), „*of*”, „*to*”, „*and*”, „*in*” and the one-letter word „*a*” fixed the frequent letters, and long partial words such as „*HrFUtoSoJF*” → „*cryptology*” revealed several rare letters at once. Human judgement is still needed.
4. **Unencrypted spaces and punctuation help the attacker:** word boundaries and lengths, and patterns like „*...'s*”, are visible directly in the cryptogram.
5. **The recovered key is an affine cipher:** c = (7p + 19) mod 26, so the whole key is described by two numbers.

6 Source Code Repository

The complete source code is available in the GitHub repository:
https://github.com/bogdan-arama/cs-labs/tree/main/lab2